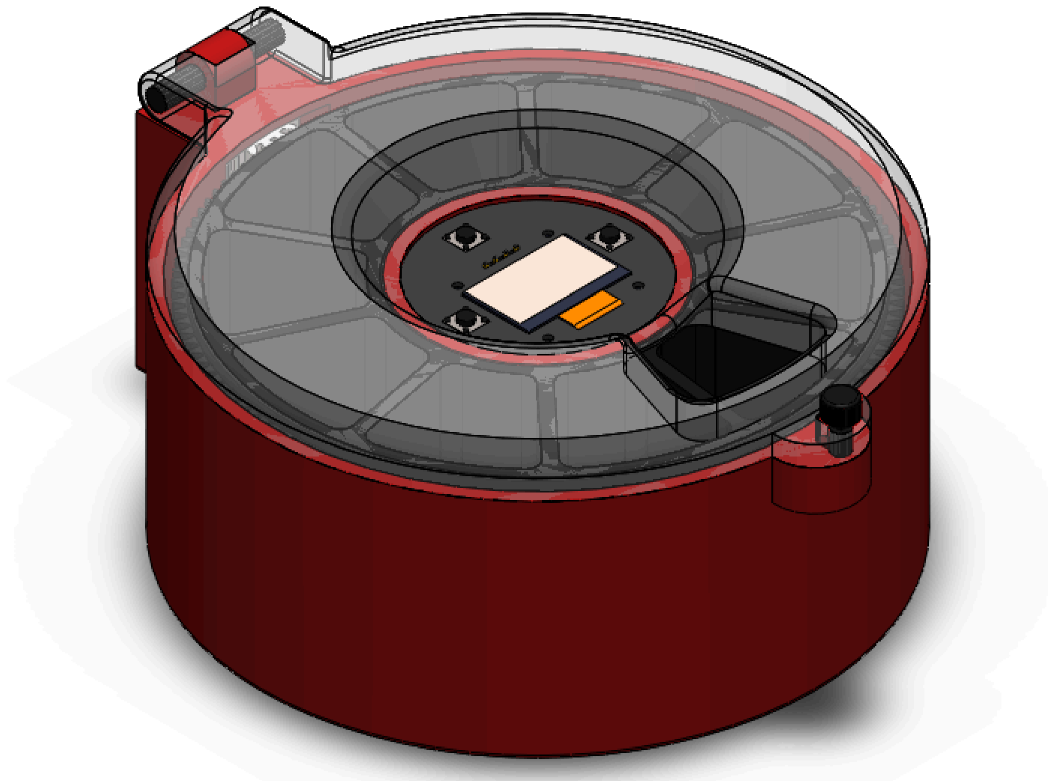

Medication Dispenser

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Prepared for: ENGS-241 Computer-Aided Design

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Abstract

The Medication Dispenser machine is an automated solution designed to remind patients and the elderly about taking medication on time and optimize the treatment plan. The Medication Dispenser machine allows its users to not only take the medication on time, but also simplifies the process for searching the right pill for patients with several cognitive disabilities.

The Medication Dispenser machine uses special preprogrammed reminders to rotate and land the inner circular tray on the right pocket. Afterwards it releases a special sound signal to let the patient know it is time to take the medications, and the only thing left to do for the patient is to come closer and take the medication. This automation reduces the need for manual intervention, allowing patients to focus on other critical tasks. The Medication Dispenser machine also includes a user-friendly interface for real-time monitoring and feedback, enabling users to track health related trends, performance and make informed decisions.

Problem Statement and Objectives

Problem Statement

Many individuals, especially the elderly or those with specific cases, often forget to take their doses at the correct times due to memory limitations, busy routines, or complex multi-dose prescriptions. Some traditional methods such as handwritten reminders, mobile alarms, or verbal instructions are often not so reliable. Furthermore, when multiple medications are assigned, patients may often consume the wrong type or incorrect dosage, increasing the risks of side effects.

Objectives

The primary objective of this machine is to ensure proper and accurate medication delivery by automating the dispensing process. The machine aims to enhance user safety and support effective treatment. Another key objective is to provide trends about user health statistics such as dispensing logs and alerts, which can inform some health providers when a dose is taken or missed. Moreover, the project seems to create a user-friendly interface while maintaining reliability and portability. The system must include customizable medication schedules to suit different treatment plans, ensure practicality and be eco-friendly.

Analysis of Existing Solutions

Current medication management solutions range from simple organizers to advanced automated dispensing systems, each with its own unique limitations. To illustrate, there are so-called basic pill organizers that come with weekly or monthly pill boxes, they help their users to separate their medication but there are no any reminder indexes for the patient to consume the medication on time. These are often cheap, but the lack of automation does not meet the requirements for some specific patients for some specific circumstances. On the other hand, there exist more advanced smart dispensers including features like dispensing, alarms, locking systems, even manual pill dosage regulation. While these decisions are effective, they are often very unaffordable for specific patient groups. Because of these limitations from both sides we have decided to invent something in the middle that would balance those features in a more simplified form and be more affordable.

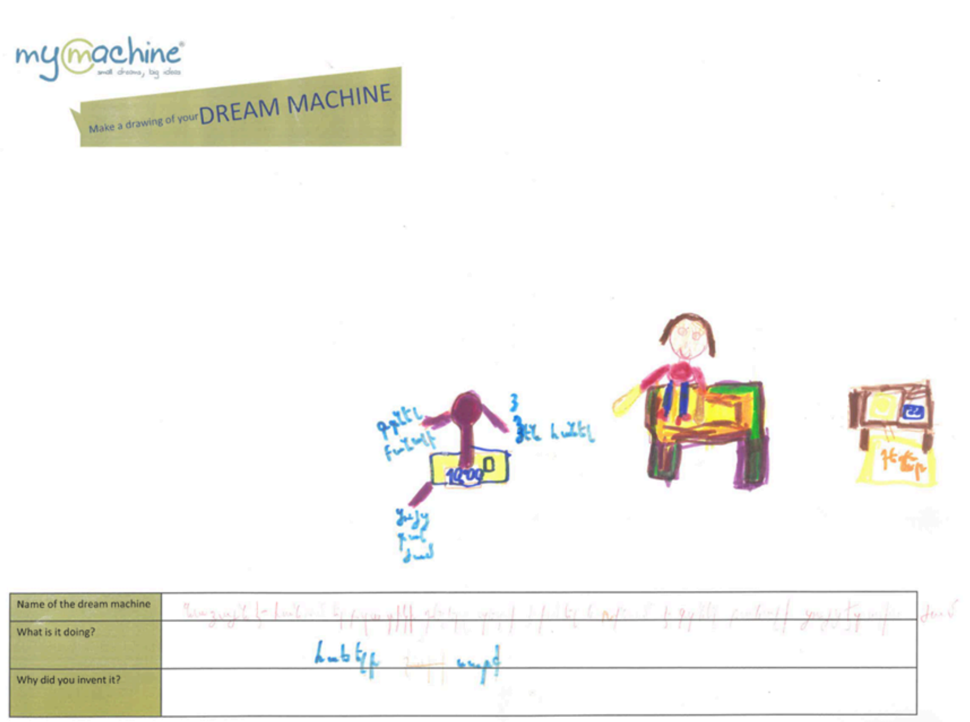
System Requirements & Constraints

The machine must be capable of reliably delivering the sorted pills in compartments at correct preset times, supported by a simple and user-friendly interface so that users or health authorities can program medication schedules. It must provide clear reminders through indexes such as light or sound, and prevent users from accessing multiple preset doses once to ensure safety. Moreover, at the same time it must be reusable and allow stakeholders to recycle it. As for now one of the important constraints in this project is the choice of material, to ensure the materials don’t react with medication inside and make it dangerous for consumption. On the other hand, the other important constraint is the weight and overall dimensions of the system. There are small advanced electronics in the market that can fit in the system and make it light and easy to carry. However, not all of them are designated for the problem we are trying to solve. Some really small ones cost a fortune, while the more affordable ones are chunky and not so easily integrable in the system.

Idea Generation & Concept Development

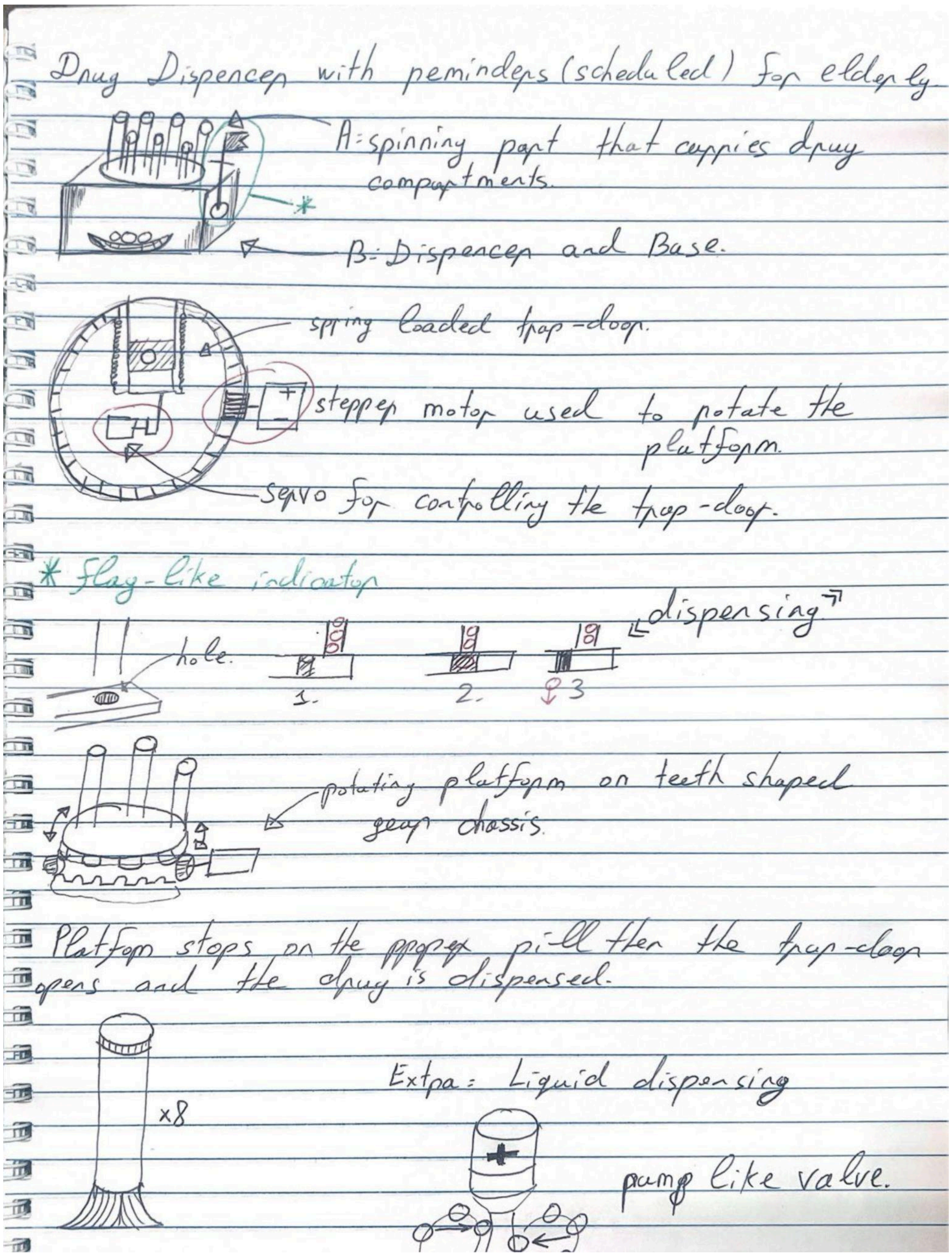
Idea Generation

The idea for our medicine dispenser originated from the “MyMachine” concept, where children sketch their dream machines and university students attempt to make those dreams into reality. Initially the child had a vision about a complex machine that would bring the medications directly to the patient. During brainstorming, we discussed how to make the idea into reality, and decided to simplify it to a device that would spin towards the proper medication and alert the patient to come and consume it. Below you can see the child’s initial sketch of the project.



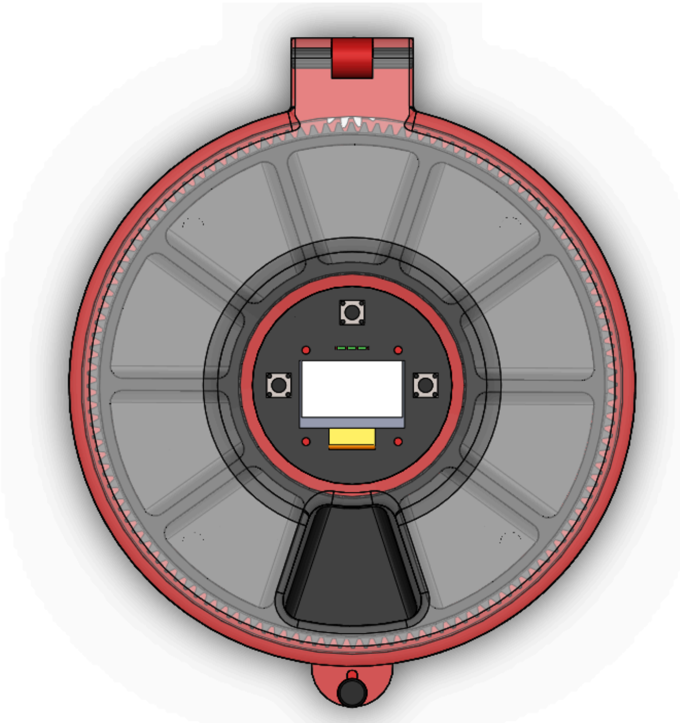
Concept Development

Below you can see our first draft of the design from our concept development stage. It is apparent that we have changed the whole design midway through the development stage.

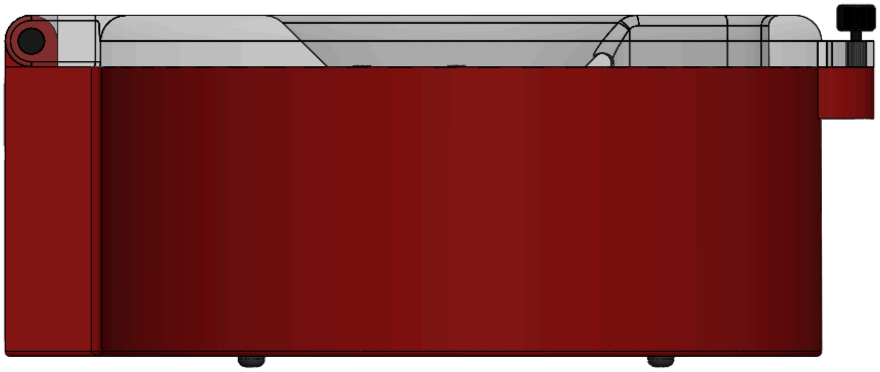


Final Design Description

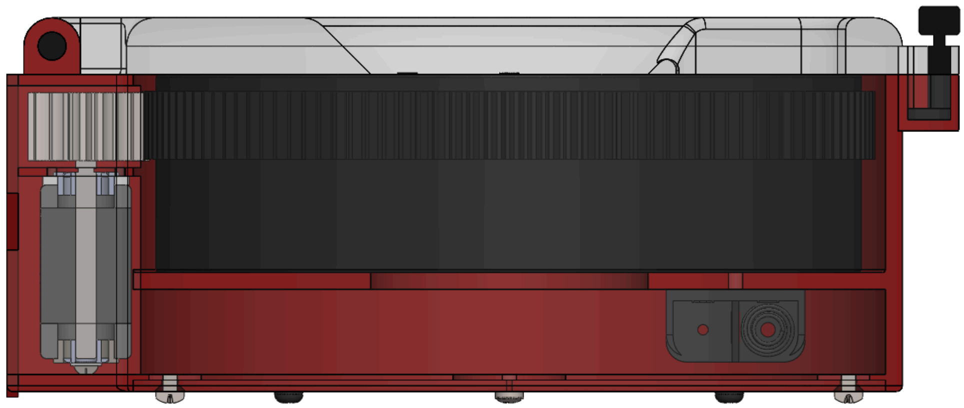
Top View:



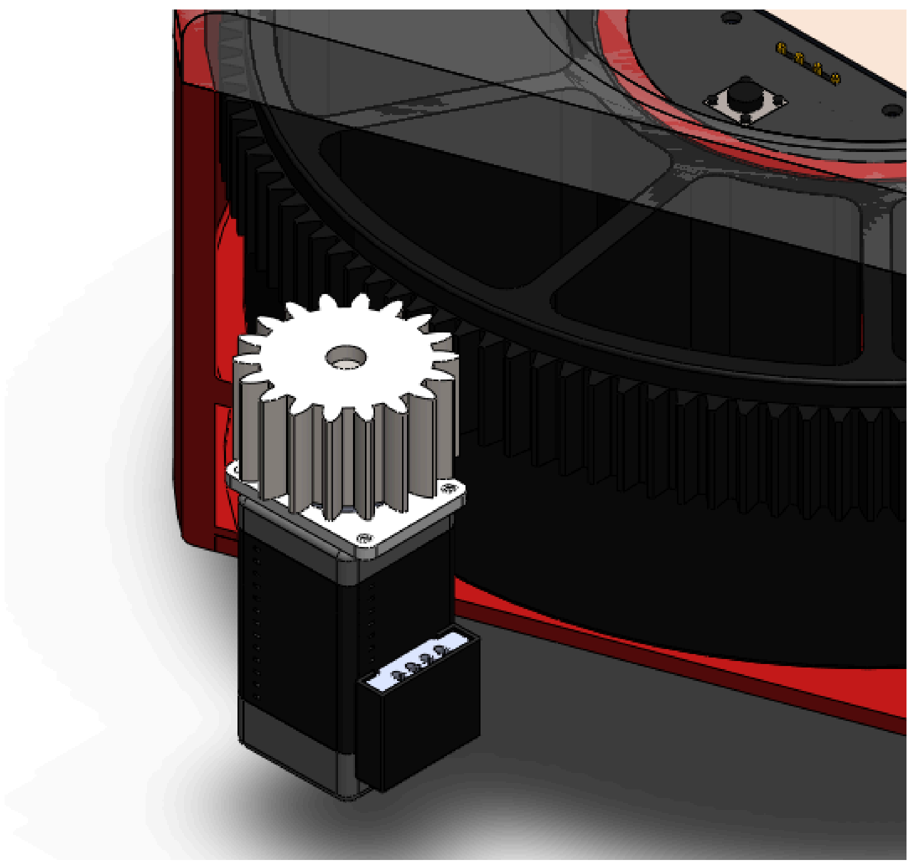
Side View:



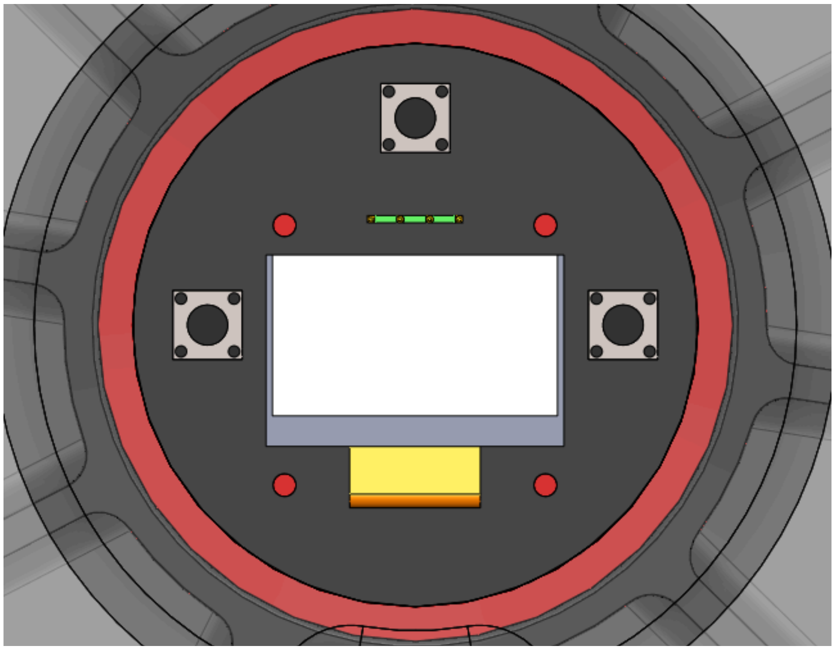
Side View (Transparent Shell):



Gear View:

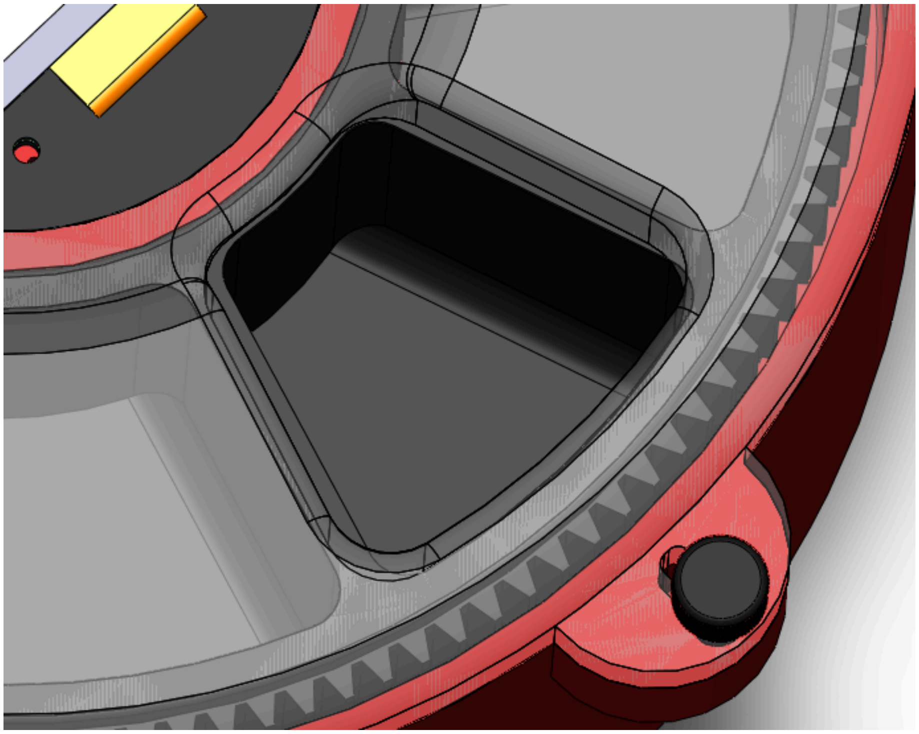


End-User Interface:

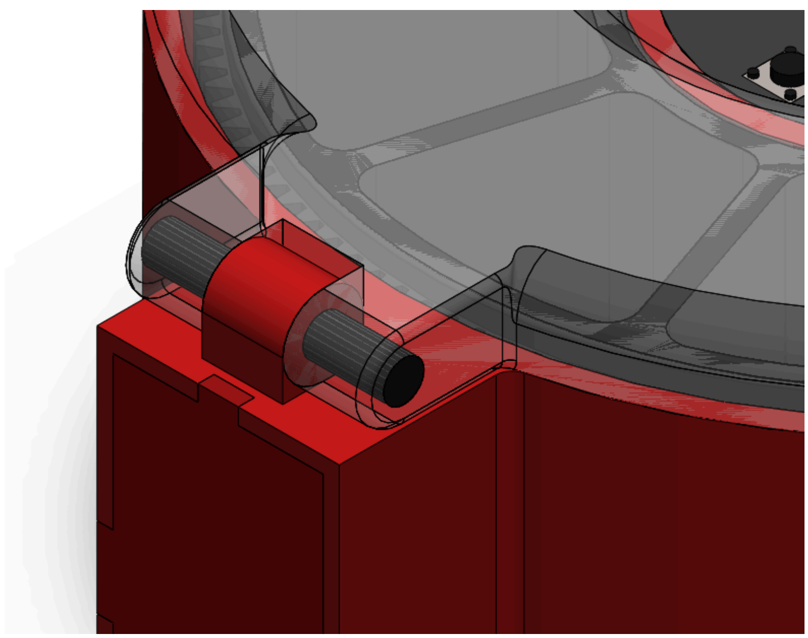


Safety Lock:

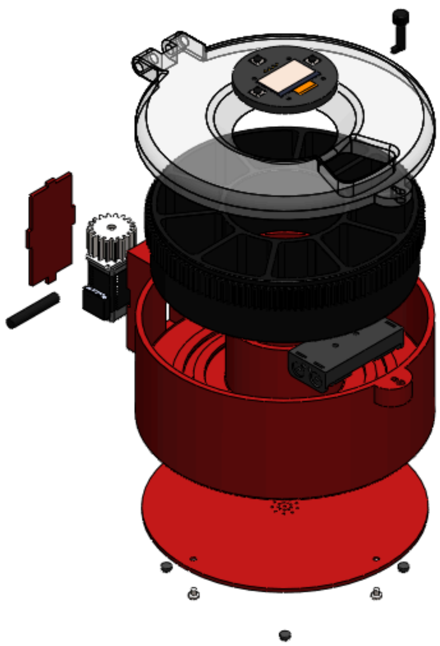
Designated Medication Pockets:



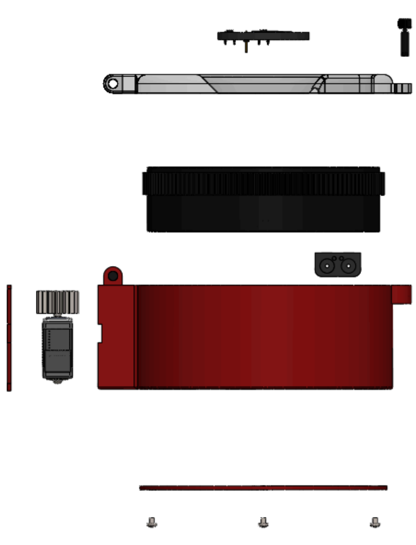
Hinged Lid Design:



Exploded View 1:



Exploded View 2:



Modeling Approach in SolidWorks

The modeling process of our device in SolidWorks began with individually creating each subcomponent of the machine. We started creating individual parts such as the rotating tray, the base shell where the rotating tray sits, and the lid. After working on these three we started implementing the stepper motor into the design. At first we decided on placing the motor in the center. Ofc it would save a lot of space, however later it would be difficult to implement the information display and buttons, so we came up with inserting the motor to the side of the rotating drum, and implement the user end interface in the center of the drum. One clever idea was not mating the stepper motor to the rotating drum, but to link those two with gears. In the future it will provide some necessary torque to make sure the patient does not rotate the drum accidentally and mix other designated hours dosage. The radial symmetry of the machine made things even more easier. You basically work on one face intersection of the machine and the program revolves it for you. All the components were created by sketching only one face intersection and revolving it around the center. After finishing the development stage and combining everything in one main assembly, it was time for tools from SolidWorks Evaluation and Simulation categories. With the help of Interference tool we quickly found the overlapping parts and fixed them, making sure that no two components rub into each other and limit the overall machine's functionality. Last, but not least, the motion study tool helped us to better represent the final design of the machine.

Simulation / Motion Study Results

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https://youtu.be/HyGgQTb41Zc?si=HtBsgTQ9gHGulyJl

Materials, BOM

The Materials selected for the machine include lightweight medium gloss plastics for the Shell, Lid, Hinge Pin, Cover, Bottom and Side covers, Battery Case. Medium gloss Rubber for Rubber Legs. Titanium for Stepper Motor Gear, and Polypropylene for the Rotating Wheel. You can see the BOM table attached below:

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Shell		1
2	Rotary Wheel With Gear		1
3	Lid		1
4	Hinge Pin		1
5	Stepper Motor		1
6	OLED TEMU 0.96 in	PART-OLED TEMU 0.96 in-DESC	1
7	Push_Button v3		3
8	Cover		1
9	Key		1
10	Battery Case		1
11	Bottom Lid		1
12	Rubber Legs		4
13	Stepper Motor Gear		1
14	Side Cover		1
15	B18.6.7M - M2.5 x 0.45 x 3 Type I Cross Recessed PHMS --3N		4

Conclusion

In this Project, we designed a medication dispenser that is simple to use, easy to build, and affordable for certain patient groups. The initial project idea was taken from the “MyMachine Armenia 2025” concept then implemented into mechanical design using SolidWorks. Throughout the project we looked at already existing products and identified their constraints. Eventually we came up with a design that balances both cost and practicality. Overall, this project helped us understand the full process of implementing basic sketches into real models, and improved our computer aided design skills.

References

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